

MEGAN TJANDRASUWITA

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EDUCATION

Massachusetts Institute of Technology
Ph.D. in Computer Science

2022 - Present

Massachusetts Institute of Technology
M.S. in Computer Science, GPA 5.0/5.0

2022 - 2024

California Institute of Technology
B.S. in Computer Science, GPA: 4.0/4.0

2018 - 2022

RESEARCH INTERESTS

Analyzing and enhancing the emergence of human-like perception and reasoning in multimodal AI. Multimodal AI for behavior analysis and social reasoning. Machine learning methods that combine deep learning with symbolic knowledge.

PUBLICATIONS

- **Megan Tjandrasuwita**, Chanakya Ekbote, Liu Ziyin, Paul Pu Liang. **Understanding the Emergence of Multimodal Representation Alignment**. *In submission to ICML 2025, ICLR 2025 Workshop on Representational Alignment (oral)*. [\[paper\]](#)
- Hengzhi Li, **Megan Tjandrasuwita**, Yi R. Fung, Armando Solar-Lezama, Paul Pu Liang. **Towards Socially-Intelligent Nonverbal Foundation Models**. *In submission to ACL 2025*. [\[paper\]](#)
- **Megan Tjandrasuwita**, Jie Xu, Armando Solar-Lezama, Wojciech Matusik. **MeMo: Meaningful, Modular Controllers via Noise Injection**. *NeurIPS 2024*. [\[paper\]](#)
- Liane Makatura, Michael Foshey, ..., **Megan Tjandrasuwita** (7th author), ... **How Can Large Language Models Help Humans in Design and Manufacturing?** *Harvard Data Science Review 2024*. [\[part1\]](#), [\[part2\]](#)
- Tailin Wu, **Megan Tjandrasuwita**, Zhengxuan Wu, Xuelin Yang, Kevin Liu, Rok Sasic, Jure Leskovec. **ZeroC: A Neuro-Symbolic Model for Zero-shot Concept Recognition and Acquisition at Inference Time**. *NeurIPS 2022, ICML 2022 Beyond Bayes Workshop*. [\[paper\]](#)
- Jennifer J. Sun*, **Megan Tjandrasuwita***, Atharva Sehgal*, Armando Solar-Lezama, Swarat Chaudhuri, Yisong Yue, Omar Costilla-Reyes. **Neurosymbolic Programming for Science**. *NeurIPS 2022 AI4Science Workshop*. [\[paper\]](#)
- **Megan Tjandrasuwita**, Jennifer J. Sun, Ann Kennedy, Swarat Chaudhuri, Yisong Yue. **Interpreting Expert Annotation Differences in Animal Behavior**. *CVPR 2021 CV4Animals Workshop*. [\[paper\]](#)
- Jasmina Arifovic, Anil Donmez, John Ledyard, **Megan Tjandrasuwita**. **Individual Evolutionary Learning and Zero-Intelligence in the Continuous Double Auction**. *Handbook of Experimental Finance, Chapter 19, p.225 – p.250, Edward Elgar publishing*

RESEARCH EXPERIENCE

Multisensory Intelligence Lab
Deep Learning Researcher

Oct 2024 - Present

Advisor: Professor Paul Liang

- Develop pipeline to train MLP and Transformer models on different modalities, including image, text, audio.
- Extensively evaluate the emergence of alignment between unimodal encoders across varying levels of redundant information and heterogeneity between modalities.

- Experiments demonstrate a nuanced understanding of when cross-modal alignment predicts model performance. Full conference paper in submission to *ICML 2025*.
- Curate social reasoning benchmark of mime performances without explicit language cues.
- Experiments show that large vision-language models struggle to infer social cues from fine-grained expression and gestures. Full conference paper in submission to *ACL 2025*.

Computer-Aided Programming Group

Sep 2022 - Present

Deep Learning Researcher

Advisor: Professor Armando Solar-Lezama

- Researched learning reusable, modular controllers for robots using a novel training objective.
- Developed ML pipeline involving imitation learning for pretraining neural network modules and reinforcement learning for transferring modules to different structures and tasks.
- Experiments in locomotion and grasping domains demonstrate that the learned modules outperform graph neural network and Transformer baselines in training efficiency.
- Full conference paper published at *NeurIPS 2024*.

Stanford Network Analysis Project, Stanford

Jun 2021 - Jun 2022

Deep Learning Researcher

Advisor: Professor Jure Leskovec

- Researched neurosymbolic methods for performing human-like concept recognition and reasoning in visual domains.
- Constructed policy architecture of agent that performs unsupervised object discovery and identifies relevant relations between concepts. Integrated recurrence in graph neural network (GNN) architecture to improve agent's long-term reasoning.
- Full conference paper published at *NeurIPS 2022*.

Machine Learning Group, Caltech

Sep 2020 - Jun 2022

Machine Learning Researcher

Advisor: Professor Yisong Yue

- Employed program synthesis to generate human-interpretable programs that classify animal behavior based on laboratory datasets.
- First-authored paper published at the *CVPR 2021 CV4Animals Workshop*.

INDUSTRY EXPERIENCE

Oracle - Corporate Architecture

Jun 2020 - Sep 2020

Software Engineer Intern

- Applied machine learning algorithms to create a recommendation system that processes purchase requests from Oracle's employees.
- Achieved highly interpretable outputs through its ranking system and had a precision of over 80% on validation examples.

TEACHING EXPERIENCE

Caltech Teaching Assistant

Jan 2022 - Jun 2022

- Held office hours, graded problem sets, and prepared lecture notes for graduate-level computer science courses: "Machine Learning and Data Mining" (CS/EE 155) and "Advanced Machine Learning Methods" (CS/EE 159).

HONORS & AWARDS

NSF Graduate Research Fellowship (GRFP)	2022
MIT Stata Family Presidential Fellowship	2022